

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Classical Dynamics and Special Relativity
L-T-P: 3-1-0 **Credits:** 4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Central force problem, Scattering in central field, Small oscillations, Rigid body motion, The inertia tensor, Eulerian angles, Euler's equation, Motion in a non-inertial frame of reference. (16)
Elementary ideas of dynamical systems. Examples in one, two and three dimensions. Fixed points and their stability and evolution of dynamical systems in phase space. Motion of a nonlinear pendulum. Tent map and elementary ideas of chaos. (12)
Michelson-Morley experiment, Lorentz transformation, Length contraction, Time dilation, Proper time, Doppler effect, Mass-energy relation. (14)

Reference Books:

1. Classical Mechanics :J.R. Taylor
2. Nonlinear dynamics and Chaos S. H. Strogatz
3. Introduction to Special relativity : R. Resnick
4. Relativity: Special, General, and Cosmological: W. Rindler

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Central force problem, Scattering in central field, Small oscillations, Rigid body motion, The inertia tensor, Eulerian angles, Euler's equation, Motion in a non-inertial frame of reference.	16
2.	Elementary ideas of dynamical systems. Examples in one, two and three dimensions. Fixed points and their stability and evolution of dynamical systems in phase space. Motion of a nonlinear pendulum. Tent map and elementary ideas of chaos.	12
3.	Michelson-Morley experiment, Lorentz transformation, Length contraction, Time dilation, Proper time, Doppler effect, Mass-energy relation.	14
Total number of hours		42